



United International University (UIU)
Department of Computer Science and Engineering (CSE)
Course: VLSI Design (CSE 4327) | Trimester: Summer 2026
Group Project - Sample Ideas and Instructions

Sample Project Ideas

Projects are to be done group-wise, based on the lab experiments covered in this course (Layout Design excluded).

No.	Sample Project Title	Idea Elaboration (What to Do)
1	CMOS Standard Cell Library: INV, NAND, NOR, AND, OR, XOR	Design transistor-level schematics and symbols of each gate in Virtuoso, verify every truth table with ADE L, and measure the propagation delay of each gate.
2	Effect of PMOS/NMOS Sizing on CMOS Inverter Performance	Sweep the Wp/Wn ratio of an inverter, measure the switching threshold, rise/fall time, and delay from DC and transient analysis, and justify the best ratio.
3	Transistor-Level Full Adder and 4-bit Ripple-Carry Adder in Virtuoso	Build a 1-bit full adder at transistor level, chain four copies using symbols, verify the sum and carry waveforms, and find the worst-case carry delay.
4	Transistor-Level 2:1 and 4:1 Multiplexer, Cross-Checked with Verilog	Design the multiplexers in Virtuoso, write the same functions in Verilog, and compare the two sets of waveforms for identical inputs.
5	Transistor-Level D Latch and D Flip-Flop: Clock-to-Q Delay and Setup Time Analysis	Build a D latch and a master-slave D flip-flop at transistor level, then measure the clock-to-Q delay and the minimum setup time from simulation.
6	Shift Register and LFSR-Based Pseudo-Random Generator in Verilog	Write a D flip-flop based shift register and an LFSR with feedback taps, simulate them, and explain the generated sequence and its length.
7	Programmable 4-bit Up/Down Counter: Simulation and Synthesis	Write a counter with up/down control and reset in Verilog, verify it with a test bench, then synthesize it and report timing, power, and area.
8	Impact of Clock Period Constraints on Synthesis Results (Area, Power, Timing Slack) of a Counter or Adder	Synthesize one counter or adder repeatedly with different clock periods, tabulate area, power, and slack, and explain the trade-offs.
9	Adder Architecture Comparison: Ripple-Carry vs. Carry-Lookahead vs. Pipelined 32-bit Adder	Write the three 32-bit adders in Verilog, verify each, synthesize them under the same constraints, and compare area, slack, and power.
10	Complete RTL-to-GDSII Flow of a Small Digital Block (e.g., 8-bit Counter or 4-bit Adder)	Take a small block through simulation, synthesis, and physical design (floorplan, power plan, placement, CTS, routing, verification) and export its GDSII file.

Note: The titles above are sample ideas only. Based on them, your group may propose a different, relevant project, but it must be approved by the instructor before you start working on it.

Project Report

Along with the project files, each group must submit a project report that includes:

- **Motivation:** why the project was chosen and what problem or concept it addresses.
- **Methodology:** the design approach, tools used, and the step-by-step flow you followed.
- **Relevant window screenshots:** for example schematics, Verilog code, test benches, waveforms or timing diagrams, and tool reports, each with a caption.
- **Results and conclusion:** what you observed and what you learned.

Submission Instructions

- Put all the project files (schematics, Verilog source, test benches, scripts, reports, and any other files used) together with the project report in a single **zip** file.
- Submit the zip file through this link: <https://forms.gle/qx7B5fcjQzRvJyt6>.